# Amna Muhammad

AI Engineer — Backend Systems

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## Summary

Software Engineering undergraduate focused on AI systems, backend development, and intelligent application design. Experienced in building API-driven systems and integrating LLM-based features into real-world applications. Seeking a Backend/AI Internship to contribute to scalable AI solutions and production-level systems.

## Education

**B.E. Software Engineering** Expected 2028
NED University of Engineering & Technology, Karachi

## Core Skills

**AI Engineering:** LLM Integration, RAG Pipelines, NLP, OpenAI API
**Backend Development:** Node.js, Express.js, Flask, FastAPI, REST API Design
**Databases:** MySQL, PostgreSQL, MongoDB, SQL, Database Design
**Programming:** Python, C++, C#
**Data & ML:** Pandas, NumPy, Scikit-learn
**Frontend (Basic):** React.js, HTML, CSS, Tailwind
**Tools:** Git, Docker, n8n, UiPath
**Collaboration:** GitHub, Remote Workflows, Agile Basics

## Experience

**AI Software Engineer Intern** Wortholic
*Duration: 2 Months*

- Developed backend systems and automated workflows using Python.
- Designed and integrated REST APIs for internal tools and system operations.
- Implemented LLM-based features using OpenAI APIs for automation and intelligent responses.
- Collaborated on testing, debugging, and optimizing system performance.

## Projects

**AI Study Partner – Collaborative Learning Platform**

- Built a multi-user AI-powered study platform for document processing and interactive learning.
- Implemented document-based Q&A, quiz generation, and flashcards using LLMs.
- Developed backend services with FastAPI and integrated a React + Vite frontend.
- Designed RAG pipelines using ChromaDB and local embeddings for context-aware responses.
- Integrated Groq APIs for fast LLM inference across summarization and Q&A tasks.
- Managed structured data and application state using PostgreSQL.

**PakIrrigo – Irrigation Water Monitoring System**

- Developed backend logic for monitoring irrigation and environmental data.
- Designed data-driven modules to support analysis and efficient resource management.

**Zenith – Exoplanet Detection (Machine Learning)**

- Built machine learning models for exoplanet detection using astronomical datasets.
- Applied preprocessing, feature engineering, and model evaluation techniques.

## Achievements

- NASA Space Apps Challenge 2025 – Local Runner-Up & Global Nominee
- Project pitching experience at IBA Tech Tank
- Participant in AI/ML hackathons and competitions

## Professional Skills

Team Collaboration • Communication • Problem Solving • Leadership